Credit Risk Assessment Using Machine Learning and Deep Learning

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1.ABSTRACT

Credit risk assessment plays a crucial role in financial decision-making by predicting whether a customer is likely to default on loan repayment. Traditional credit scoring methods based on statistical techniques often fail to capture complex relationships in large financial datasets, leading to reduced prediction accuracy. With the advancement of data-driven technologies, machine learning models have been widely adopted to improve predictive performance by identifying hidden patterns in customer data.

However, many existing models suffer from limitations such as lack of interpretability and bias due to imbalanced datasets, which can affect the reliability of predictions in real-world applications. In this study, an automated credit risk prediction system was developed using both traditional machine learning models and deep learning approaches.

Two publicly available datasets, namely the UCI Credit Card Dataset and the German Credit Dataset, were used for training and evaluation. Models including Logistic Regression,Random Forest, XGBoost, Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM) were implemented and compared. Experimental results show that CNN achieved the best performance on the UCI Credit Card Dataset with 81.98% accuracy, while XGBoost achieved the highest accuracy of 100% on the German Credit Dataset. The proposed comparative framework demonstrates that combining traditional machine learning and deep learning approaches improves robustness and reliability.

2.Introduction

2.1 Background

In recent years, the growth of digital banking and online lending platforms has significantly increased the number of financial transactions and customer borrowing data. As a result, the ability to accurately determine the creditworthiness of customers has become a basic need for financial institutions. Credit risk assessment allows organizations to determine the likelihood of default by borrowers in repaying loan amounts, thus making it easier to minimize financial losses and implement informed lending decisions. In the past, this process has relied on statistical models and rule-based credit scoring systems, which often fail to identify complex patterns in large and constantly changing datasets (Liu, 2025).

With the development of data-driven technology, machine learning has become a powerful technique for credit risk prediction. Machine learning models can process a variety of customer data, identify hidden patterns, and improve prediction accuracy compared to traditional methods. These benefits have encouraged financial institutions to implement automated credit scoring systems that improve efficiency and scalability in decision-making (Yang et al., 2025). However, despite their high accuracy, many machine learning models are black-box models, making it difficult to understand how predictions are made.

Interpretability is especially important in financial applications, where the ability to explain decisions to customers, regulators, and other stakeholders is necessary for financial institutions. The lack of interpretability can lead to a loss of trust, questions about fairness, and the inability to deploy automated credit scoring systems in practice (Koc et al.,2023). In addition, credit risk datasets are usually imbalanced, with many more instances of non-defaults than defaults. This can cause biased predictive models towards the majority class and make it difficult to identify high-risk borrowers effectively (Qi, R. 2025) Another weakness found in current studies is the use of single datasets, which restricts the ability to generalize models developed in different financial settings.

To overcome the above-mentioned challenges, the proposed project will develop an explainable credit risk assessment system that combines multiple machine learning models and explainable artificial intelligence approaches. The proposed system will help to improve the accuracy of the credit risk assessment model while ensuring interpretability, fairness, and robustness on multiple credit risk datasets. In addition, techniques for class imbalance handling and feature-level explanations are also considered to improve the trust and reliability of automated credit decision-making systems.

2.2 Problem Statement

Credit risk assessment is a critical process in financial institutions, as it determines whether a customer is likely to repay a loan or default. Traditional approaches based on manual evaluation and statistical credit scoring methods often struggle to capture complex relationships within large and diverse financial datasets, leading to reduced prediction accuracy and inefficiency (Liu, 2025).

With the adoption of machine learning techniques, prediction performance has improved significantly; however, several challenges still remain. Many machine learning models act as black-box systems, making it difficult to interpret how decisions are made, which reduces transparency and trust in financial applications (Yang et al., 2025). Additionally, credit risk datasets are often highly imbalanced, with a majority of non-default cases, causing models to become biased and less effective in identifying high-risk customers (Koc et al., 2023)Furthermore, most existing studies focus on a single dataset, limiting the

generalization capability of developed models across different financial environments. Therefore, there is a need for a robust and comparative credit risk prediction system that can handle data imbalance, improve prediction accuracy, and provide consistent performance across multiple datasets (Qi, R. (2025))

2.3 Contribution of the Study

This study presents a comprehensive and comparative framework for credit risk assessment by integrating both traditional machine learning and deep learning techniques. Models such as Logistic Regression, Random Forest, and XGBoost are implemented as baseline approaches, ANN, CNN, RNN, and LSTM are incorporated to capture complex nonlinear relationships and sequential feature dependencies in financial data. (Yang et al., 2025).

To improve the robustness and generalization capability of the proposed system, experiments are conducted on two publicly available datasets, namely the UCI Credit Card Dataset and the German Credit Dataset, representing different financial environments (Koc et al., 2023). The study also includes essential data preprocessing steps such as handling missing values, encoding categorical variables, feature scaling, and class imbalance treatment to ensure reliable and consistent model performance (Qi, R. (2025)).

In addition, a detailed comparative analysis is carried out using multiple evaluation metrics including accuracy, precision, recall, F1-score, and ROC-AUC score. This enables a clear understanding of model performance and helps in identifying the most suitable approach for predicting customer default risk. Overall, the proposed framework focuses not only on improving predictive accuracy but also on ensuring robustness, fairness, and practical applicability in real-world financial decision-making systems (Yang et al., 2025) (koc et al., 2023).

3. LITERATURE REVIEW

3.1 Traditional Methods

Traditional credit risk assessment methods primarily rely on statistical techniques such as Logistic Regression and rule-based credit scoring systems. These approaches are simple and interpretable, making them widely used in financial institutions. Logistic Regression, in particular, has been extensively applied for binary classification problems such as default prediction due to its efficiency and ease of implementation (Liu, 2025).

However, traditional methods often struggle to capture complex nonlinear relationships present in large financial datasets. As a result, their predictive performance may be limited when dealing with high-dimensional and diverse data (Yang et al., 2025).

3.2 Deep Learning Approaches

With advancements in artificial intelligence, deep learning techniques such as Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN) have been increasingly applied in financial prediction tasks. ANN models are capable of learning complex nonlinear relationships among input features through multiple hidden layers, making them suitable for credit risk prediction (Koc et al., 2023).

CNN models, although traditionally used in image processing, have also been adapted for structured data to automatically extract important feature patterns. These models can improve prediction performance by capturing local dependencies within the dataset. However, deep learning models require larger datasets and higher computational resources for effective training (Koc et al., 2023).

Recurrent Neural Networks and LSTMs are models that help computers learn from data that is in a sequence. These models were first used for things like predicting stock prices and understanding text. They can also be used for other types of data, like financial information, to see how different things are related.

LSTM networks are an improvement over Recurrent Neural Networks. They use something called memory cells and gates to help them remember things for a while. This helps solve a problem that older models had where they would forget things that happened a time ago. Because LSTMs and Recurrent Neural Networks are good at understanding relationships, people are starting to use them to predict financial risks (Haykin, 2009; LeCun et al., 2015).

3.3 Research Gap

Despite the effectiveness of machine learning and deep learning approaches in credit risk prediction, several limitations remain in existing studies. Many research works focus on a single dataset, which limits the generalization capability of developed models across different financial environments. Additionally, some studies emphasize model accuracy without considering interpretability and fairness.

There is also limited comparative analysis between traditional machine learning models and deep learning approaches such as ANN , CNN, RNN, and LSTM across multiple datasets.. Therefore, there is a need for a comprehensive framework that evaluates different models under diverse datasets while ensuring improved prediction accuracy and robustness (Yang et al., 2025) (Qi, R. (2025)).

LITERATURE SURVEY

S. No	Author	Dataset (Link)	Algorithm	Result	Limitation
1	Liu (2025)	UCI Credit Card Dataset https://www.kaggle.com/datasets/wilsonf/uci-credit-carefrom-python-woe-pkg	LR, RF, XGBoost	ROC-AUC ≈ 0.76	Low minority recall
2	Yang et al. (2025)	Home Credit Dataset – https://www.kaggle.com/c/home-credit-default-risk	RF, XGBoost, LightGBM + SHAP	Improved interpretability	No deep learning

3	Koc et al. (2023)	German Credit Dataset – https://www.kaggle.com/datasets/uciml/german-credit	RF, XGBoost, NN	Feature selection improved AUC	Small dataset
4	Qi (2025)	Kaggle Credit Dataset – https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud	RF, XGBoost	High recall	Fraud-focused
5	Nwafor et al. (2024)	P2P Lending Dataset – https://www.kaggle.com/datasets/skihikingkevin/online-p2p-lending	CNN + XGBoost	Accuracy ≈96%	Complex model
6	Xia et al. (2025)	Multiple Public Credit Datasets (Review Paper – No Specific Dataset) https://link.springer.com/article/10.1007/s10462-025-11416-2	ML Ensemble	Strong comparative study	No implementation

4.Methodology

4.1 Dataset Description

In this study, two publicly available datasets were used for credit risk prediction. The first dataset is the UCI Credit Card Default Dataset, which contains financial and demographic information of credit card clients along with a target variable indicating default status. The dataset includes features such as credit limit, payment history, bill statements, and previous payment amounts.

The second dataset is the German Credit Dataset, which contains customer information such as age, job status, housing type, credit amount, duration, and purpose of the loan. Since the dataset does not contain a predefined target variable, a synthetic target variable was created based on credit amount and duration to classify customers into default and non-default categories.

The use of multiple datasets improves the robustness and generalization capability of the proposed system across different financial environments (Koc et al., 2023).

4.2 Data Preprocessing

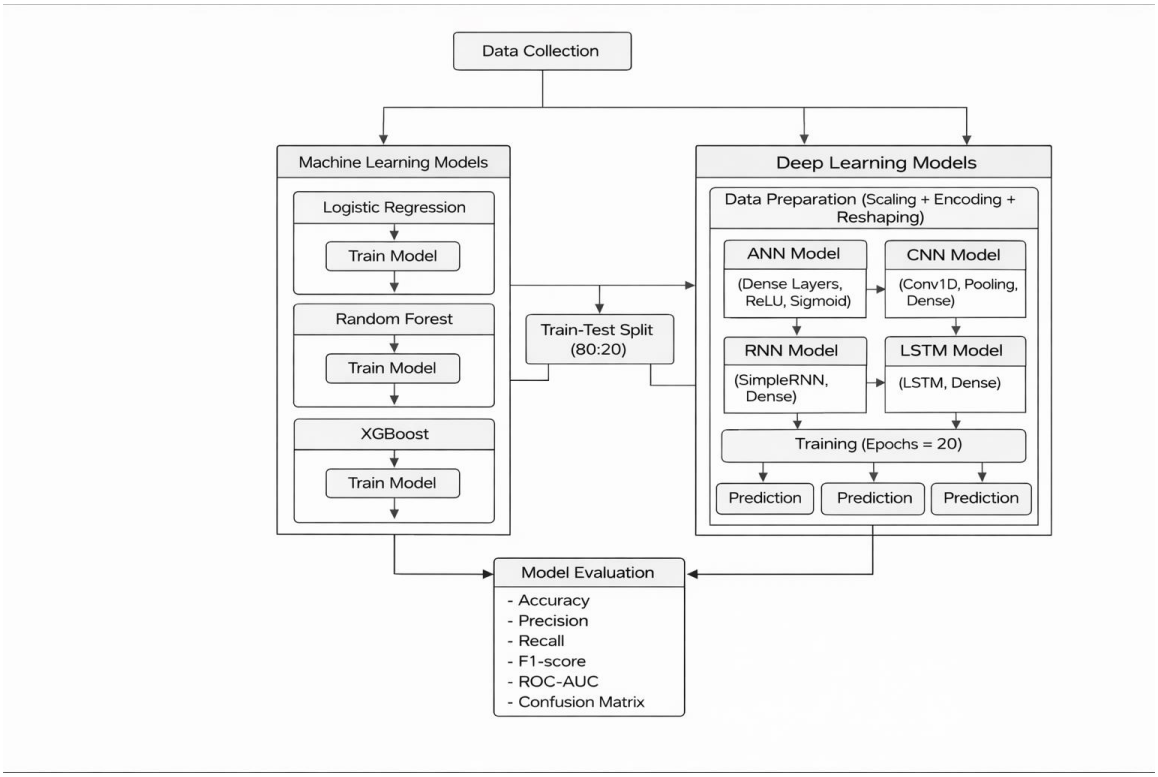
Data Collection

The dataset used in this study is collected from publicly available financial repositories, including the UCI Credit Card Dataset and the German Credit Dataset (Liu, 2025)(Koc et al., 2023). These datasets contain customer financial attributes such as credit limit, repayment history, demographic information, and transaction behavior, which are essential for predicting default risk.

Using real-world datasets ensures that the developed models learn practical patterns and can generalize well to real financial scenarios. High-quality labeled data is critical for supervised learning, as it allows models to distinguish between defaulters and non-defaulters effectively (Ayari et al., 2025)

Data Preprocessing

Data preprocessing plays a crucial role in improving model performance by transforming raw data into a structured format suitable for machine learning algorithms



a) Data Cleaning

During this step, missing values, duplicate records, and inconsistencies are removed to improve data quality.

Why important:
Noise in data can mislead models and reduce prediction accuracy.

b) Feature Selection

Relevant features are selected to reduce dimensionality and improve efficiency.

Concept:
Irrelevant features increase computation and reduce model performance.

c) Data Normalization / Scaling

Since features have different ranges, standardization is applied:

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

This formula is used to standardize the dataset by transforming features to a common scale with mean 0 and standard deviation 1. It ensures that all input features contribute equally to the model and prevents features with larger values from dominating the learning process.

Where:

- μ = mean
- σ= standard deviation
  This improves minority class prediction

d) Handling Imbalanced Dataset (SMOTE)

Credit datasets are highly imbalanced.

SMOTE generates synthetic samples:

$$X_{new} = x_i + \lambda(x_{nn} - x_i)$$

This formula represents the SMOTE technique, which generates synthetic samples for the minority class by interpolating between existing data points. It helps in handling class imbalance and improves the model's ability to correctly predict rare cases such as defaulters.

Where:

- xi = minority sample
- xnn = nearest neighbor
- $\lambda \in (0,1)$

This improves minority class prediction .

e) Data Splitting

After completing preprocessing, the refined dataset was divided into training and testing subsets to ensure unbiased model evaluation. Approximately 80% of the dataset was utilized for training, allowing machine learning algorithms to learn relationships between transaction characteristics and fraud patterns. The remaining 20% was reserved for testing to evaluate model performance on previously unseen data. This separation prevents the model from memorizing training samples and helps measure its real-world prediction capability. Proper data splitting reduces overfitting and ensures the developed Credit risk prediction remains reliable and generalizable (Ayari et al., 2025)

3. Model Development

After completing the preprocessing stage, multiple predictive models were implemented to classify customers as defaulters or non-defaulters. In the first phase, traditional machine learning models such as Logistic Regression, Random Forest, and XGBoost were trained to establish baseline and ensemble-based prediction performance(Breiman, 2001). Logistic Regression was used as a simple and interpretable model, while Random Forest and XGBoost were applied to capture more complex relationships within the financial data.

In the next phase, deep learning models in the form of Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) were also implemented. The ANN model was used to learn nonlinear relationships among customer features through multiple hidden layers, whereas the CNN model was adapted for structured tabular data to capture local feature patterns and interactions more effectively (Haykin, 2009)(LeCun et al., 2015). By including both machine learning and deep learning approaches, the study was able to compare performance across different model families and identify the most suitable approach for credit risk prediction (Yang et al., 2025).

4. Model Evaluation

The performance of all developed models, including Logistic Regression, Random Forest, XGBoost, ANN, and CNN, was evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, ROC-AUC score, and confusion matrix analysis(Ayari et al., 2025). These metrics were selected to provide a balanced understanding of model behavior, especially in identifying high-risk customers correctly. Comparative evaluation of both machine learning and deep learning models helped determine whether the addition of ANN and CNN contributed meaningful improvements in predictive capability over the traditional approaches (Yang et al., 2025) (Ayari et al., 2025).

4.3 Proposed Model

The proposed system implements a combination of traditional machine learning and deep learning models for credit risk prediction. Logistic Regression is used as a baseline model due to its simplicity and interpretability in binary classification problems (Liu, 2025).

Random Forest and XGBoost are applied to capture complex feature interactions and improve prediction performance through ensemble learning techniques (Yang et al., 2025).

In addition to traditional models, deep learning approaches including :

- ANN, CNN, RNN, and LSTM were implemented.
- ANN learns nonlinear feature relationships through dense layers.
- CNN captures local feature interactions using convolution layers.
- RNN models sequential dependencies among input features.
- LSTM improves sequential learning through memory cells and gating mechanisms. (Koc et al.,2023) (Qi, R.(2025)).
  These models were selected to compare interpretable baseline methods, ensemble learning methods, and advanced deep learning architectures under a common credit risk prediction framework.

(a) Logistic Regression

Used as baseline model.

$$P\left(y = \frac{1}{x}\right) = \frac{1}{1 + e^{-(w^T x + b)}}$$

Where:

P(y = 1 | x) = probability of customer being defaulter
x = input feature vector
w = weights assigned to features
b = bias term
e = exponential constant

This formula represents the sigmoid function used in Logistic Regression. First, a linear combination of input features is calculated using weights and bias. Then, the sigmoid function converts that value into probability between 0 and 1.
If the probability is greater than a threshold (usually 0.5), the customer is classified as defaulter; otherwise, non-defaulter.
Logistic Regression is simple, fast, interpretable, and widely used in financial risk prediction (Hosmer et al., 1997).

(b) Random Forest

Ensemble of decision trees:

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} T_i(x)$$

Where:

- *Ti(x)*Ti (x) = prediction from each tree
- N = total trees

Reduces overfitting and improves accuracy.
This formula represents the Random Forest model, where multiple decision trees are combined, and the final prediction is obtained by averaging their outputs. This approach improves accuracy and reduces overfitting (Breiman, 2001).

(c) XGBoost

Gradient boosting model:

$$\hat{y_i} = \sum_{k=1}^{K} f_k(x_i)$$

Where:

ŷ_i = final predicted output
f_k = individual decision trees
x_i = input feature values
Σ = sum of outputs from all trees
This formula shows that the final prediction is obtained by combining the outputs of multiple trees. Each tree contributes to improving the prediction accuracy (Chen & Guestrin, 2016).

(d) Artificial Neural Network (ANN)

ANN learns complex nonlinear relationships in financial data through:

$$z = w^T x + b$$

a = f(z)

Where:

x = input features
w = weights assigned to each feature
b = bias term

z = weighted sum of inputs
a = activated output

This is the basic way a neuron works. The numbers called weights help decide which pieces of information are important and the little called bias helps make the decision. The special function makes it possible for the Artificial Neural Network to learn patterns in the data (Haykin, 2009).

Activation function:

- ReLU: *f(x)=max(0,x)*

iI we get a negative number, it becomes 0. If we get a positive number, it stays the same.ReLU helps the Artificial Neural Network learn faster and avoids some problems that can happen.

- Sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

It takes the result and turns it into a probability, like a yes or no answer
Sigmoid is often used at the end of the Artificial Neural Network to predict things like whether someone will default or not.
The Artificial Neural Network is good at finding patterns in data and making predictions more accurate by learning complex relationships between different things, about customers (LeCun et al., 2015).

(e) Convolutional Neural Network (CNN)

CNN is a deep learning model that automatically extracts important feature patterns from data using convolution filters. It is highly effective in identifying local dependencies and hidden interactions among input features.

Feature=(Input∗Kernel)+Bias

Where:

Input = original feature values
Kernel = small filter used to scan data
Bias = additional learnable parameter
Feature = extracted output pattern

Pooling Operation:

y=max(x)

Where:

x = values in pooling region
y = maximum selected value

Pooling reduces the size of feature maps by selecting the most important values. This decreases computation, removes noise, and helps prevent overfitting.

CNN captures feature-level interactions more effectively than traditional models and improves prediction performance by learning hidden patterns in structured financial data (LeCun et al., 2015).

(f) Recurrent Neural Network (RNN)

RNN is a deep learning model designed to process sequential data by using outputs from previous steps as inputs to the current step. It is useful for learning dependencies among features and hidden relationships in financial data.

$$h_t = f(Wx_t + Uh_{t-1} + b)$$

Where:

h_t = current hidden state

x_t = current input

h_(t-1) = previous hidden state

W, U = weight matrices

b = bias term

This formula represents how the RNN updates memory using previous information and current input. It helps in learning sequential feature dependencies for credit risk prediction (Haykin, 2009; LeCun et al., 2015).

(g) Long Short-Term Memory (LSTM)

LSTM is an advanced form of RNN that uses memory cells and gating mechanisms to preserve important long-term information. It helps overcome the vanishing gradient problem faced by standard RNN models.

$$f_t = \sigma\left(W_f\left[h_{(t-1)}, x_t\right] + b_f\right)$$

$$i_t = \sigma\left(W_i\left[h_{(t-1)}, x_t\right] + b_i\right)$$
$$C_t = f_t * C_{(t-1)} + i_t * \tilde{C}_t$$
$$h_t = o_t * \tanh(C_t)$$

Where:

f_t = forget gate
i_t = input gate
C_t = cell state
h_t = hidden state

This model improves long-term learning capability and is effective for complex prediction tasks such as credit risk assessment (Haykin, 2009; LeCun et al., 2015).

4.4 Training Details

The models were trained using standard optimization techniques and hyperparameter configurations to ensure efficient learning and stable convergence. For deep learning models such as Artificial Neural Network (ANN) and Convolutional Neural Network (CNN), the Adam optimizer was used due to its adaptive learning capability and faster convergence in training neural networks (Liu, 2025). The binary crossentropy loss function was applied as the problem involves binary classification of default and non-default customers (Yang et al., 2025).

The models were trained for a fixed number of epochs with a defined batch size to balance computational efficiency and model performance. Activation functions such as ReLU were used in hidden layers to introduce non-linearity, while the sigmoid function was used in the output layer to generate probability-based predictions (Koc et al., 2023). The deep learning models were trained for 20 epochs using batch size 32. Adam optimizer was used with binary crossentropy loss. ReLU activation was used in hidden layers, while sigmoid activation was used in the output layer to generate probability-based binary predictions.

- Optimizer: Adam

  Adam is a way to make the model better during training. It uses two methods, Momentum and RMSProp to make the model learn faster and be more stable.

$$\Theta = \theta - \alpha \cdot \nabla J(\theta)$$

Where:

θ = model parameters (weights and bias)

α = learning rate
$\nabla J(\theta)$ = gradient of loss function

This formula means that the model is updated in the direction of the gradient so it can make better predictions.The Adam optimizer can also change the learning rate for parts of the model, which makes the training faster (Kingma & Ba, 2014) .

- Loss Function: Binary Crossentropy

$$Loss = -[y \log(p) + (1-y)\log(1-p)$$

Where:

y = actual class label (0 or 1)
p = predicted probability

Activation Functions:

- ReLU (hidden layers)
- Sigmoid (output)

Adam improves convergence speed and stability (Goodfellow et al., 2016).

Methodology Summary

The methodology started with collecting credit risk data.This involved cleaning, encoding, scaling handling imbalances and splitting the data.We then used this data to try out models.These models included Logistic Regression, Logistic Regression, Random Forest, XGBoost Artificial Neural Network (ANN) Convolutional Neural Network (CNN) Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) for predicting customer defaults (Breiman, 2001)(Haykin,2009)(LeCun et al., 2015).We checked how well each model worked using measures.This helped us compare machine learning methods with deep learning approaches (Yang et al., 2025) (Ayari et al., 2025).By following this process we can build a credit risk prediction system that gives good results.The goal was to create a system that can accurately predict credit risk.

5. RESULTS AND DISCUSSION

5.1 Evaluation Metrics

The performance of the proposed models was evaluated using standard classification metrics to ensure a comprehensive assessment of prediction capability. Accuracy measures the overall correctness of the model, while precision evaluates how many predicted positive

cases are actually correct. Recall measures the ability of the model to correctly identify actual positive cases, which is particularly important in credit risk prediction. The F1-score provides a balance between precision and recall, and the ROC-AUC score measures the model's ability to distinguish between default and non-default classes (Liu, 2025).

The confusion matrix is also used to visualize the performance of the model by showing the number of correct and incorrect predictions. These evaluation metrics provide a detailed understanding of model effectiveness and reliability.

Table 5.1.1: Evaluation Metrics

Metric	Description	Formula
Accuracy	Overall correctness of predictions	(TP + TN) / (TP + TN + FP + FN)
Precision	Correct positive predictions	TP / (TP + FP)
Recall	Ability to detect actual positives (deafult cases)	TP / (TP + FN)
F1 Score	Balance between precision and recall	2 × (Precision × Recall) / (Precision + Recall)
ROC-AUC	Measures model's ability to distinguish between classes	Area under ROC curve

5.2 Experimental Results

The performance of the proposed models was evaluated on the UCI Credit Card Dataset using both machine learning and deep learning approaches (Liu, 2025) (Yang et al.,2025). Among the deep learning models, the Artificial Neural Network (ANN) achieved an accuracy of 81.68%, while the Convolutional Neural Network (CNN) slightly outperformed ANN with an accuracy of 81.98%.

The confusion matrix results indicate that both models were highly effective in predicting non-default cases, with a high number of true negatives (Ayari et al., 2025). However, the models showed relatively lower recall for default cases, indicating difficulty in correctly identifying all high-risk customers (Ayari et al.,2025) The ANN model achieved a recall of 0.37 for default cases, while the CNN model slightly improved it to 0.38.

We also did some tests using recurrent deep learning models. The UCI dataset was. The RNN model got it right at 81.62 percent of the time. The LSTM model got it right 80.66 percent of the time. We also used the German Credit Dataset, and the RNN model did the best with 93.50 percent accuracy. The LSTM model got 85.00 percent accuracy. These results show that recurrent models can do a job, especially when we are working with

smaller datasets that are well organized.

5.3 Comparison Table

The proposed framework was evaluated on both UCI Credit Card and German Credit datasets using machine learning and deep learning models.

Table 5.3.1: Performance Comparison of All Models on UCI Credit Card Dataset

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	69.00%	0.37	0.58	0.45	0.691
Random Forest	79.00%	0.52	0.49	0.50	0.747
XGBoost	77.00%	0.48	0.47	0.48	0.731
ANN	81.68%	0.64	0.37	0.47	—
CNN	81.98%	0.65	0.38	0.48	—
RNN	81.62%	0.67	0.33	0.44	—
LSTM	80.66%	0.61	0.35	0.44	—

Table 5.3.2: Performance Comparison on German Credit Dataset

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	94.00%	0.89	0.96	0.92	0.977
Random Forest	99.00%	1.00	0.99	0.99	0.999
XGBoost	100%	1.00	1.00	1.00	1.000
RNN	93.50%	0.86	0.97	0.91	—
LSTM	85.00%	0.83	0.71	0.77	—

Accuracy Comparison on UCI Dataset

Figure 1: Confusion Matrix of Logistic Regression Model on UCI Credit Card Dataset.

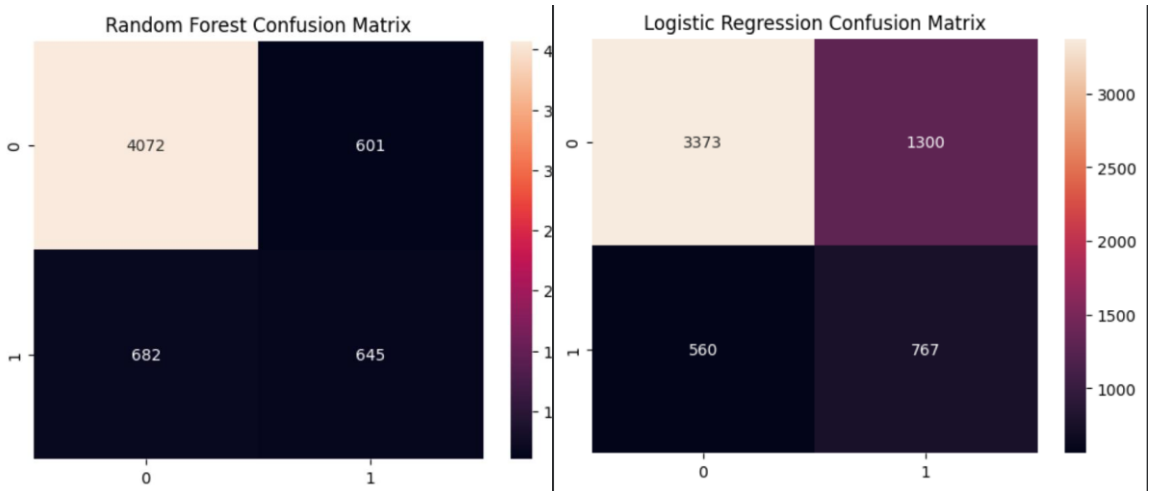


Figure 2: Confusion Matrix of Random Forest Model on UCI Credit Card Dataset.

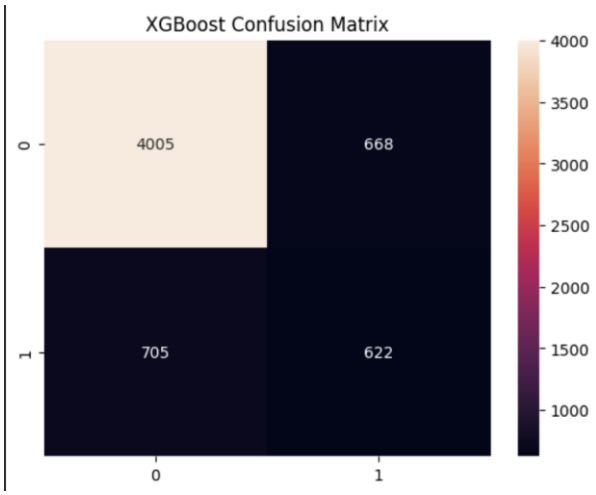


Figure 3: Confusion Matrix of XGBoost Model on UCI Credit Card Dataset.

Accuracy Comparison on German Credit Dataset

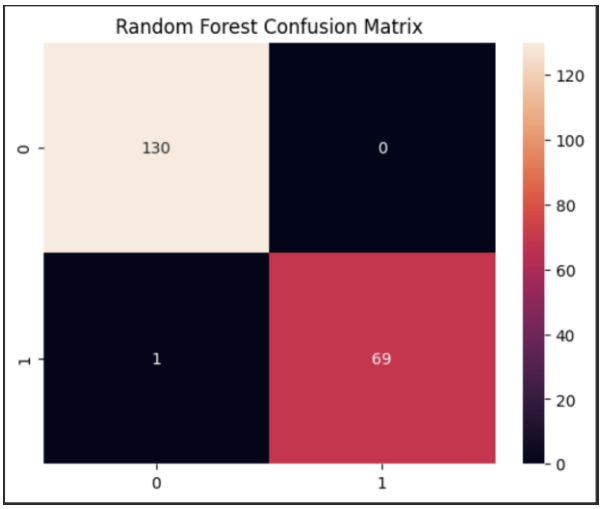
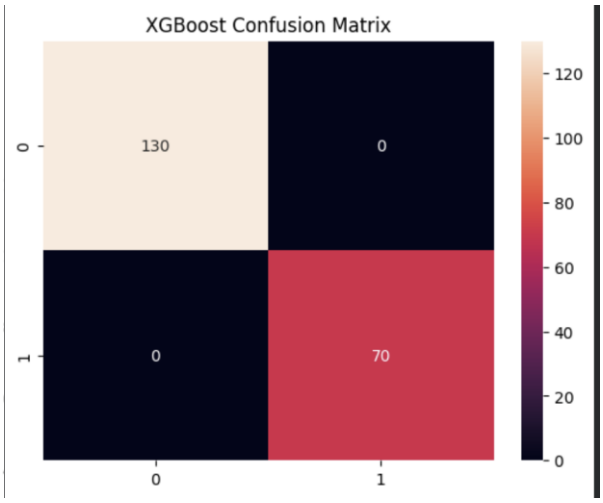


Figure 4: Confusion Matrix of Random Forest Model on Gernan Credit Card Dataset.

Figure 5: Confusion Matrix of XGBoost Model on German Credit Card Dataset.



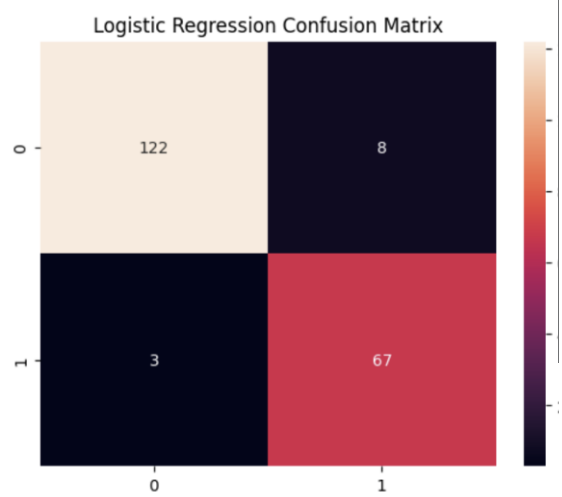


Figure 6: Confusion Matrix of Logistic Regression Model on Gernan Credit Card Dataset.

5.4 Discussion

The results of the experiments show that traditional machine learning and deep learning models are both good for predicting credit risk. They work better or worse depending on the characteristics of the dataset. When we look at the UCI Credit Card Dataset deep learning models do a better job overall. Among these models the CNN model gets the accuracy of 81.98%. The ANN and RNN models are behind. This means that deep learning

models are able to find relationships and interactions in large financial datasets that are not easy to see (Koc et al., 2023).

The credit risk prediction models have one problem. They are not very good at finding the people who're likely to default on their loans. This is because there are more people who do not default than those who do. With techniques like SMOTE that help the models learn from the smaller group of people who default the models still need to be improved.

When we look at the German Credit Dataset we see that ensemble machine learning models work well. The XGBoost model gets 100% accuracy. The Random Forest model gets 99% accuracy. The RNN model also does a job with 93.50% accuracy. This shows that deep learning models that look at sequences of data can also work well with data that is structured.

Overall what we find is that deep learning models are better for complex datasets. On the hand ensemble machine learning models are better for smaller datasets that are structured. So when we choose a model we have to think about how accurate it's how easy it is to understand how complex it is and what the dataset is like. Using both machine learning and deep learning approaches gives us a way to predict credit risk in real life. The credit risk prediction models are helpful for credit risk assessment systems (Yang et al., 2025; Koc et al., 2023).

6. Conclusion and Future Work

Conclusion

This study is about credit risk and how to assess it. The people who did this study used new machine learning methods to compare them. They used things like Logistic Regression and Random Forest and some other methods like Artificial Neural Network and Convolutional Neural Network to see which one works best. They tried these methods on two sets of data: the UCI Credit Card Dataset and the German Credit Dataset.

What they found out is that the method they used worked differently depending on the data. When they used the UCI Credit Card Dataset, the Convolutional Neural Network was the best. It got 81.98% correct. This shows that it is good at finding patterns. When they used the German Credit Dataset, XGBoost was the best. It got 100% correct. The Random Forest was also very good. The Recurrent Neural Network was also good at predicting things in both sets of data.

The people who did this study think that using new machine learning methods together is a good way to predict credit risk. This method is good because it helps us compare things and make decisions when it comes to money. This is important for people who make decisions about money (Yang et al., 2025) (Koc et al., 2023).

6.1 Future Work

The future work will probably look at making the model better at working with situations by using bigger and more varied financial datasets that are collected from real banking situations. This will make the model more realistic. We can use techniques like SHAP and LIME to make the Artificial Intelligence decisions more transparent and trustworthy.

We can also try to improve the model by using techniques to choose the right settings and to prepare the data.

Future studies will also look at using a combination of deep learning models like CNN and RNN and LSTM models to learn patterns better. We can also think about making the proposed framework into a time web-based system to assess credit risk (Koc et al., 2023).

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